Project #1

Data Visualization LECD 2024/25

Introduction

The main goal of **project #1** of the course is developing and applying a set of skills that allow effective decision-making on the choice and application of a visualization model for a given task(s) and a dataset. The students should perform a careful selection of multiple existing visualizations that in **their opinion** most effectively represent the dataset at hand. A written critical reflection essay should support all the decisions. It is also intended that the students acquire and solidify their knowledge about the main concepts of the programming languages and frameworks that will be used throughout the course. In particular, Plotly, Python, and Pandas. The ultimate goal of the first project is the acquisition of important skills, such as analysis and synthesis, organization and planning, critical thinking, application of knowledge in practice, investigation, and autonomous learning. **This set of skills will be crucial for successfully completing the main project of the course!**

Objectives

The objectives of the project are the following:

1. Choose a dataset and decide what attributes will be used. Any data cleaning and filtering are welcome, but not mandatory.
2. Select a set of **4 visualization techniques** that in the opinion of the students are most appropriate for representing the chosen data. This goal implies a survey of the **existing** methods, which can be done using the Internet, the recommended literature, and scientific articles, among other sources of information. It is highly recommended to ask for guidance from your teachers. Remember, not all the visualization techniques are implemented in the Plotly framework.
3. Apply all 4 of the chosen techniques to visualize the chosen dataset using the Plotly framework. **The same attributes must be used in all 4 visualizations.** The students should be using **Graph Objects** as the main tool to construct and customize visualizations, thus it is discouraged to use Express in the first project.
4. Write a **critical reflection** essay about all 4 visualizations and the decision taken. This should not exceed **2-3 paragraphs** of running text. As a starting point try to answer these questions:
    a. Which visualization provides clearer insight into data, and what aspects made you think that?

b. Which visualization is more memorable?
c. What aspect of the data can you recall and which of the 4 visualizations comes to your mind first?
d. Which visualization technique is better for communication and which is for exploration/analysis tasks, and why?
e. Which tasks can you identify that can be accomplished with each visualization and how well it supports that task? Among other questions.

Groups

There are no groups. The project should be developed individually.

Evaluation

The grading for the project is equal to 25% of the final grade, which is equivalent to 5 out of 20.

Important notice! It is not possible to improve the grade of the project.
Another important notice! The submissions made after the deadline will not be accepted.

Grading criteria

The following elements will be evaluated:

- Visualization selection: Was your survey extensive enough or did you pick the first graphics seen on the Internet? Was at least one visualization model an effective solution? Are the sources of information well cited and credited?
- Visual quality: Are your solutions visually appealing? Do the visualizations you created follow good design practices?
- Implementation: Quality of the implementation (Is your code robust? Does your code follow conventions and overall good programming practices?)
- Crítical reflection: Is your reflection concise and well-reasoned or superficial and vague? Does it develop around several aspects of work or focus on a single theme?

Minimum grade

To get approval for the Data Visualization course, it is necessary to obtain a classification equal to, at least, 40% of the final grade of the project.

Data

The students are given a google drive repository with multiple datasets to choose from. It is recommended to focus on one dataset and on the attributes that are important for the main idea. Remember, the first project is not about the data.

Deliverables

The students should submit one single Jupiter notebook that should contain all the required materials: i) a brief description of the dataset and the attributes; ii) the code for each visualization; iii) the critical reflection; iv) the references. For that, the students are provided with this Jupiter template, which structure should not be modified. The students can additionally submit the PDF of the rendered version of the notebook to avoid any problems with the rendering on a different computer.

Important notice! The submission deadline should be consulted on the General Information page of the course on the Inforestudante platform.

Dissemination

Projects developed in the context of this course may be submitted to be displayed in FeedNPlay, a multimedia device with a total width of more than 6 meters, which is installed on the 1st Floor of the Department of Informatics Engineering. Students should follow the instructions available at the FeedNPlay website (https://feednplay.dei.uc.pt).

Plagiarism and fraud policy

According to the Article 28.º of the disciplinar reglament for the UC students (https://www.uc.pt/regulamentos/ga/vigentes/Regulamento_Disciplinar_Estudantes_UC)

"2 — A fraude cometida em sede de avaliação de uma unidade curricular implica a anulação imediata dessa avaliação e leva à reprovação liminar do estudante na inscrição na unidade curricular em causa, devendo ser registada na plataforma informática da UC e averbada no processo individual do estudante."